AI-based Neural Data Science

BME447, BME547

Instructor

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Teaching Assistant

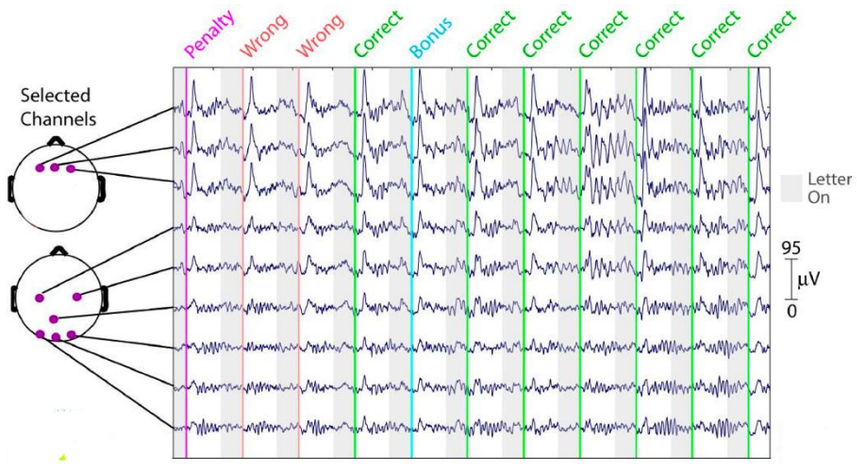
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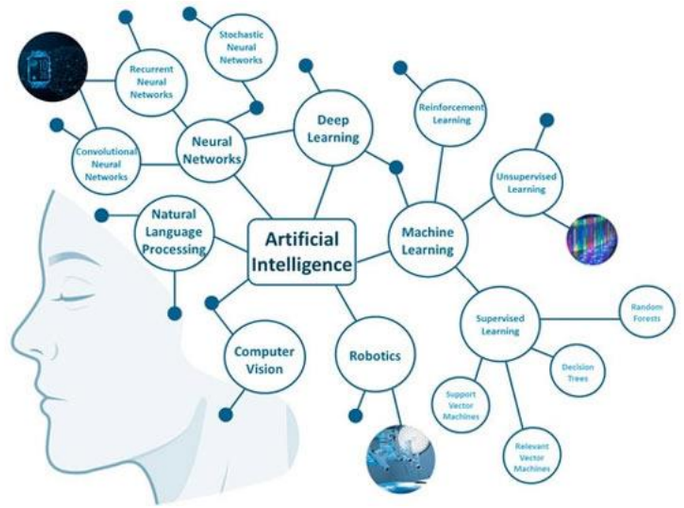
- Bldg. #104, room 903

Aims

Neural Data



AI Methods



Decode information about brain functions

Predict behavior from neural signals

Understand how the brain works

Neural Data Analysis

- Students will practice the analysis of neural data
  - Team-based analysis
  - AI assistance is allowed if the history of prompts is submitted
- Neural data used in the class
  - Open data (OpenNeuro)
  - Multiple tasks (Simon, Oddball, Interval Timing, Resting)
  - Parkinson’s disease vs. Healthy control
  - Detailed information – Next week

Schedule

Week	Topic
1	Introduction to Neural Data Science
2	Project Overview
3	Characteristics of EEG
4	Characteristics of EEG
5	Holiday
6	Holiday
7	Unsupervised Learning
8	Midterm break

Schedule

Week	Topic
9	EEG Analysis I
10	Supervised Learning
11	EEG Analysis II
12	Deep Learning
13	EEG Analysis III
14	Brain Networks
15	EEG Analysis IV
16	Final Presentation

Project

- Team
  - Recommendation: maximum 4 students/team
  - Team size doesn’t affect evaluation
    - Separate teams between graduate and undergraduate
- Project topic
  - Neural data analysis using the AI methods
    - No need to use all the methods taught in class
  - Every team will use the same data

Project

- Team evaluation
  - Competition among teams
  - Key: Scientific/Engineering merits
    - E.g. Novel findings / technical advances / …
  - Project outcomes from multiple teams could be combined together for publication
- In-class presentation
  - Each team will present their on-going results in the class
  - Unique results and insights will be highly appreciated

Attendance check

- Attendance will be checked by calling attendance in the class
- It will start from 7 Sept.

Grading

- (Individual) Attendance: 20%
- (Group) Project: 70%
- (Group) Peer Evaluation: 10%